

DETAILED ESTIMATE

Name of the Road :- Construction of Aamli Fala Bridge and Road

Block : Dungarpur

Length of the Road (Km) :

4.00 KM

District : DUNGARPUR

PART - " N " - Retaining Wall & Bed Protection

700.00 M

PART - N		Description of Item	No	L (m)	B (m)	D (m)	Qty	Unit
		2		4	5	6	7	8
1	1	Earthwork in excavation for foundation of structure complete with all <u>lift, dewatering, shoring and shuttering</u> etc. Including refilling of trenches in 250 mm layer, ramming, watering and disposal of surplus soil <u>within a lead of 1000 meters</u> , as per drawing and technical specification. [MoRTH Specification : Clause 304]. in ordinary soil [MoRTH Specification : clause 301] BY MECHANICAL MEANS Depth upto 3m	1	700.00	1.60	1.50	1680.00 cum 1680.00 cum	
2	3-2	Plain Cement Concrete <u>1:3:6</u> with 40mm nominal mix in foundation with crushed stone aggregate 40mm nominal size mechanically mixed, placed in foundation and <u>compacted by vibrator including curing for 14 days</u> . (MoRTH specification clause : 2100)	1	700.00	1.60	0.15	167.99 cum 167.99 cum	
3		Providing & laying mechanically mixed and vibrated Cement Concrete in foundation upto a depth of 1.5mtr, below ground level and 1.5mtr. above ground/bed level. [MoRTH Specification : Clause 1000/1700]. (a) Cement concrete 1:3:6 mix with crushed stone aggregate 20 mm nominal size.	1	700.00	$\frac{0.45}{2}$	$\frac{1.3}{2}$	2.50	1531.25 cum 1531.25 cum
4	B-3	Providing A.C. pipe Weep holes 150mm dia in Masonry/plain Concrete/Reinforced Concrete abutment, wing wall/return wall complete as per drawing and technical specification. [MoRTH Specification : Clause 2205].	120	1.00	-	-		120.00 Mtr

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P.W.D. Sub. En. Jhonthri

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HYDRAULIC DESIGN

Construction of Aamli Fala Bridge and Road

Design of 1000 mm dia RCC H. P. Culvert

Catchment area

By Dickens Formula

$$\begin{aligned} Q &= \text{CM}^{3/4} \\ &= 14 \quad 0.1188^{3/4} \\ &= 14 \quad 0.202354 \\ &= 2.83296 \text{ cumecs} \end{aligned}$$

Discharge which can pass through 1000 mm dia pipe
Flowing full with operating head

1.00 m.

$$Q = y \sqrt{2gh}$$

Length of the pipe = 7.50 M
Diameter = 1.00 M
(From IRC Paper No.13 Plate-16/page176
for circular pipe with rounded Entry
For L= 7.50 M and D = 1.00 M
The Conveyance Factor $y = 0.686$

$$\begin{aligned} Q &= y \sqrt{2gh} \\ &= 0.686 \sqrt{2 \times 9.8 \times 1.00} \\ &= 3.037051 \text{ cumecs} \end{aligned}$$

$$\text{No. Of Pipes} = \frac{2.83296}{3.037051} = \boxed{0.932799} \text{ Nos.}$$

Say 1 Nos. Rows

Hence provide 1000mm dia R.C.C. Hume pipe 1 Row

Mahendra Das

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HYDRAULIC DESIGN

Design of 1000 mm dia RCC H. P. Culvert

Catchment area = 0.32 Sqkm. [0.400 x 0.800]

By Dickens Formula

$$Q = CM^{3/4}$$

$$= 14 \quad 0.32^{3/4}$$

$$= 14 \quad 0.425464$$

$$= 5.956491 \text{ cumecs}$$

Discharge which can pass through 1000 mm dia pipe
Flowing full with operating head 1.00 m.

$$Q = \frac{y\sqrt{2g}h}{\dots}$$

Length of the pipe = 7.50 M
Diameter = 1.00 M
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for circular pipe with rounded Entry
For L = 7.50 M and D = 1.00 M
The Conveyance Factor y = 0.686

$$Q = \frac{y\sqrt{2g}h}{\dots}$$

$$= \frac{0.686 \sqrt{2 \times 9.8 \times 1.00}}{\dots}$$

$$= 3.037051 \text{ cumecs.}$$

$$\text{No. Of Pipes} = \frac{5.956491}{3.037051} = \boxed{1.96127} \text{ Nos.}$$

Say 2 Nos. Rows

Hence provide 1000mm dia R.C.C. Hume pipe 2 Row

Mahendra Kote
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